COURSE DESCRIPTION			
Program	**Architecture, Interior Design**		
Course Title	**Masonry & Service Lab**		
Course Code	**2187**	**Semester**	**II**
Type of Course	**Lab**	**Contact Hours**	**45**
Teaching Scheme (L: T:P:R)	**0:0:3:0**	**Credits**	**1.5**
CIE Marks	**60**	**ESE Marks**	**40**

Introduction

This laboratory course provides hands-on training in masonry construction practices and basic building services essential for architectural execution.

Course Objectives

1	To identify the suitability of various materials, and understand their relevant characteristics.
2	Setting out buildings from drawings and familiarize soil types
3	To supervise the masonry construction
4	To perform plumbing works in buildings and familiarize tiles used for construction

Course Pre-requisite

Topic	Course code	Course Title	Semester
Knowledge of basic Mathematics	1002	Fundamentals of Engineering Mathematics	I
Knowledge about basic Physics	2002A		I
Basic knowledge about Building materials	2182	Building Materials & Technology	II

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Identify the different building materials and conduct tests to find out their strength related to construction activities	Apply
CO2	Set out the building at site from the given building plan and familiarization of different types of soil.	Apply
CO3	Construct and supervise brick masonry works	Apply
CO4	Perform basic plumbing works and familiarization of tiles used for construction.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-		3	-	-	-	3	-	3	-
CO2	3	-	-	-	2	-	3	3	-	3	-
CO3	3	-	-	-	-	-	2	3	-	3	-
CO4	3	-	-	-	-	-	2	3	-	3	-
Total	12	-	-	3	2	-	7	12	-	12	-
Correlation Level	3	-	-	3	2	-	2.33	3	-	3	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
1	0	2	0	2.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Determine the fineness of cement	Assess cement quality	CO1	Apply	1
2	Determine the standard consistency of cement	Identify the amount of water required	CO1	Apply	2
3	Determine the initial and final setting time of cement	Interpret the setting behaviour of cement	CO1	Apply	3
4	Compression test on bricks	Determine the compression strength	CO1	Apply	1
5	Tensile test on mild steel rod (Demonstration)	Interpret strength parameters	CO1	Apply	2
6	Determination of fineness modulus of coarse and fine aggregates	Classify aggregates	CO1	Apply	6
Module II					
7	Setting out of the building from the given plan	Execute site layout	CO2	Apply	3
8	Identification of soil types at foundation pits (Site visit and report)	Relate soil to foundation	CO2	Apply	3
	Series Test - I		CO2	Apply	3
Module III					
9	Construct one brick thick wall in English bond	Construct masonry bond	CO3	Apply	1.5
10	Construct one and half brick thick wall in English bond	Execute masonry work	CO3	Apply	1.5
11	Construct one brick thick wall in Flemish bond	Construct masonry bond	CO3	Apply	1.5
12	Construct one and half brick thick wall in Flemish bond	Execute masonry work	CO3	Apply	1.5
			CO1	Remember	
Module IV					
13	Identify different types of flooring tiles and prepare report about the specifications	Identification of flooring tiles	CO4	Apply	3
14	Plumbing work for tap and shower	Install plumbing fixtures	CO4	Apply	3
15	Plumbing work for water closet	Install sanitary fittings	CO4	Apply	3
	Series Test - II		CO4	Apply	3

Open-ended experiment

Objective: To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.
Description: At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.
Students may choose to: Conduct field tests on cement sample and comment on the properties. Construct a two brick thick wall English and Flemish bond.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Identification of building materials.	2
2	Visit a nearby construction site and identify the construction materials.	3
3	Interpretation of material test results.	2

4	Virtual lab test on tension test on steel.	2
	Module I	
5	Prepare a layout of a two room building for setting out.	3
6	Identify the type of foundation suitable for the site you visited. Justify your answer.	2
	Module I	
7	Sketch masonry bonds and construction details	3
8	Identify the bond used in construction site in your locality and prepare a sketch.	3
9	Prepare a layout of bond used for walls in your locality.	3
10	Find out the modifications in masonry when other types of bricks(concrete brick, interlock brick etc.) used for masonry.	2
	Module I	
11	Prepare a plumbing layout of your home	3
12	Prepare the electrical layout of your room	3
13	Prepare a plumbing layout for a given plan	3
14	Prepare the electrical layout for a given plan	3
	Total Self-Learning hours	37.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Concrete Technology	Shetty M.S.	S. Chand and Co. Pvt. Ltd., Ram Nagar, Delhi.
2	Concrete Technology	Gambhir M.L.	Tata McGraw Hill Publishing Co. Ltd., Delhi.

REFERENCE			
Sl. No.	Title	Author	Publication
1	Concrete Technology	Santhakumar A R	Oxford University Press, New Delhi.
2	Laboratory Manual in Concrete Technology	Sood, H., Kulkarni P. D., Mittal L. N	CBS Publishers,New Delhi.

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://ms-nitk.vlabs.ac.in/List%20of%20experiments.html	I
2	https://cs-iitd.vlabs.ac.in/exp/cube-test/	I
3	https://eerc01-iiith.vlabs.ac.in/List%20of%20experiments.html	I
4	https://nptel.ac.in/courses/105102012	I
5	https://www.youtube.com/watch?v=7dN3Iku0Bns	II
6	https://www.youtube.com/watch?v=gTsosoYEfgM&list=PLyEuOm4kr6Ceh_B7UNUDJvIVZGXDR56GL	III

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Vicat’s apparatus	IS: 5513-1996	1
2	IS sieves		1
3	Universal Testing Machine		1
4	Masonry tools set		1 set
5	Electrical & plumbing kits		1 set

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity

Assessment Methodology – Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50